

MINAHIL MOBIN

MS Cybersecurity Student • AI & Cybersecurity • Machine Learning • Generative AI • Cloud Security

GitHub • LinkedIn • Portfolio

Education

National University of Computer and Emerging Sciences (FAST-NUCES) **Lahore, Pakistan**

MS Cyber Security (In Progress) August 2025 - June 2027

◦ Relevant Coursework: Advanced Network Security, Applied Information Security, Machine Learning for Cybersecurity, Research Methodology

National University of Computer and Emerging Sciences (FAST-NUCES) **Lahore, Pakistan**

BS Data Science August 2021 - June 2025

◦ Relevant Coursework: Machine Learning, Data Mining, Artificial Intelligence, Database Systems, Data Structures and Algorithms, Software Engineering

Experience

AI Engineering Intern, Abacus July 2026 - Present

- Developed Python applications using object-oriented programming, file handling, exception handling, and modular design.
- Built automation tools for document processing, text analysis, and JSON report generation.
- Performed data cleaning, exploratory data analysis (EDA), and visualization using Pandas and Matplotlib.
- Implemented and evaluated machine learning models including Linear Regression, Logistic Regression, Decision Trees, and K-Means using Scikit-learn.
- Built an AI Knowledge Assistant using LLM API integration, prompt engineering, embeddings, ChromaDB, semantic search, RAG-based document question answering, and a FastAPI backend with a basic AI agent workflow.

MIS Intern, Shaukat Khanum Memorial Cancer Hospital June 2024 - July 2024

- Explored applications of Large Language Models (LLMs) in healthcare, including clinical information retrieval, documentation assistance, and decision support.
- Reviewed research papers, technical reports, and case studies to identify emerging trends in Medical Information Systems and healthcare AI.
- Evaluated publicly available healthcare datasets for their suitability in predictive analytics and machine learning applications.

Academic Projects

Explainable AI-Based Intrusion Detection System for Cloud Security May 2026

Technologies: Python, Scikit-learn, Pandas, NumPy, SHAP, Random Forest, Decision Tree, SVM, Flask, AWS EC2

- Developed a machine learning-based Intrusion Detection System (IDS) for cloud security using the CICIDS2017 dataset and evaluated Decision Tree, Random Forest, and Support Vector Machine (SVM) classifiers.
- Performed data preprocessing, feature selection, label encoding, feature scaling, and model evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis.
- Selected Random Forest as the best-performing model and deployed it as a Flask REST API on an AWS EC2 instance for real-time intrusion detection.
- Conducted cross-file, cross-dataset, and noise robustness evaluations to assess model performance on unseen and perturbed network traffic without retraining.
- Integrated SHAP (SHapley Additive exPlanations) to explain model predictions and identify the network traffic features that most influenced intrusion detection decisions.
- Demonstrated the limitations of model generalization across different datasets, highlighting the importance of robust cross-dataset validation for practical cloud deployment.

Feature Selection-Based Lightweight Intrusion Detection System May 2026

Technologies: Python, Scikit-learn, Pandas, NumPy, Random Forest, Mutual Information, CICIDS2017

- Designed and implemented a lightweight Random Forest-based Intrusion Detection System (IDS) using

Mutual Information feature selection on the CICIDS2017 intrusion detection dataset.

- Preprocessed network traffic data through cleaning, feature selection, normalization, and binary class labeling before model training and evaluation.
- Experimentally evaluated seven feature subsets (5, 10, 15, 20, 25, 30, and 40 features) to determine the minimum effective feature set for intrusion detection.
- Achieved 99.82% classification accuracy using only 15 of 78 network traffic features, reducing the feature space by 80.8% while maintaining high detection performance.
- Analyzed the trade-off between feature reduction, model complexity, and detection performance using Mutual Information-based feature importance analysis.

Dyslexia Support Hub (AI Learning Platform)

August 2024 - May 2025

Technologies: React, Django, Python, PostgreSQL, Machine Learning

- Designed and developed an AI-assisted educational platform to support children with dyslexia through personalized and adaptive learning experiences.
- Built adaptive learning modules that dynamically adjusted content difficulty based on individual student performance and learning progress.
- Integrated machine learning models to analyze student performance and generate personalized learning recommendations.
- Developed a full-stack web application using React for the frontend and Django for the backend, implementing RESTful APIs for seamless communication.
- Implemented secure user authentication, progress tracking, interactive learning activities, and performance dashboards.
- Collaborated throughout the software development lifecycle, including system design, implementation, testing, debugging, and technical documentation.

Real Estate Price Prediction System

May 2024

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib

- Developed a machine learning-based regression model to predict residential property prices using historical housing datasets.
- Performed exploratory data analysis (EDA), data preprocessing, missing value handling, outlier detection, and feature engineering to improve model performance.
- Trained and evaluated multiple regression models using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score.
- Visualized feature relationships, data distributions, and prediction results to support model interpretation and performance analysis.
- Implemented an end-to-end machine learning pipeline using Python, Pandas, NumPy, and Scikit-learn.

Inventory Management System

February 2023 - May 2023

Technologies: React, Django, Python, PostgreSQL

- Developed a full-stack inventory management system for managing products, inventory levels, and stock transactions.
- Designed responsive user interfaces using React and implemented backend services with Django and Python.
- Implemented CRUD operations, input validation, and database integration to ensure efficient inventory management.
- Integrated user authentication and role-based access control to secure application functionality.
- Participated in project planning, testing, debugging, and technical documentation throughout the software development lifecycle.

School Management System

April 2023 - May 2023

Technologies: PHP, MySQL, HTML, CSS

- Developed a web-based School Management System for managing students, teachers, attendance, and academic records.
- Designed and implemented a relational MySQL database to efficiently store and retrieve institutional data.
- Implemented role-based authentication and authorization for administrators, teachers, and students.
- Developed modules for student registration, attendance management, and academic record maintenance.
- Optimized SQL queries and validated user input to improve database performance and application reliability.

Candy Crush Clone

December 2022

Technologies: Assembly Language

- Developed a simplified Candy Crush-style game using Assembly Language by implementing core gameplay mechanics.
- Implemented tile matching, score calculation, board updates, and game logic using low-level programming techniques.
- Applied debugging and optimization techniques to improve execution efficiency and memory utilization.
- Strengthened understanding of computer architecture, registers, interrupts, and memory management through practical implementation.

Pac-Man Clone

May 2022

Technologies: C++, Object-Oriented Programming

- Developed a Pac-Man-style game in C++ using object-oriented programming principles and modular software design.
- Implemented player movement, enemy AI, collision detection, score tracking, and game state management.
- Designed reusable classes and modular components to improve code maintainability and scalability.
- Tested, debugged, and optimized gameplay mechanics to ensure smooth and reliable user interaction.

Technical Skills

Programming Languages: Python, JavaScript, SQL, PHP, C++, Assembly

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Classification, Regression, Model Evaluation, Explainable AI (SHAP)

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings

Web Development: React, Django, REST APIs, JSON

Cybersecurity: Intrusion Detection Systems (IDS), Network Security, Cloud Security, Security Analytics, Wireshark, Weka

Cloud & Databases: AWS EC2, MySQL, PostgreSQL

Software Engineering: Object-Oriented Programming (OOP), Git, Software Testing, Debugging

Certifications

AWS Academy Graduate - Cloud Security Builder	May 2026
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AWS Academy Graduate - Cloud Security Foundations	April 2026
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AWS Academy Graduate - Cloud Foundations	April 2026
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